

CS 137 HW 2 Report

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April 2026

Abstract

Accurate weather forecasting remains one of the most computationally demanding challenges in atmospheric science. While deep learning offers a data-driven alternative to numerical fluid dynamics, deploying Convolutional Neural Networks (CNNs) on meteorological data often yields "black box" models. In this work, we introduce SkyOracle, a forecasting framework designed to predict local weather conditions 24 hours in advance over the New England region. We explicitly detail the training of two distinct architectural paradigms: a standard multi-task CNN and a residual network utilizing Global Average Pooling (GAP). Furthermore, we rigorously evaluate the networks' physical intuition using stratified spatial saliency mapping to verify their reliance on actual atmospheric phenomena. Finally, we present an independent ablation study investigating how the introduction of an auxiliary extreme-weather classification task impacts continuous representation learning in the residual backbone.

1 Model Architectures and Training Protocols

1.1 Data Formulation and Processing

Our objective is to forecast local weather conditions at the Tufts University Jumbo Statue grid point with a 24-hour lead time. The input, X_t , is a multi-channel spatial snapshot spanning a 450×449 Lambert conformal grid. At any given time t , the model ingests a 42-channel spatial tensor to predict conditions 24 hours later ($t + 24h$).

Because meteorological features exhibit extreme variance in physical scale, we preprocessed all 42 input channels using channel-wise z-score normalization derived exclusively from the training distribution. To ensure training stability, we systematically imputed expected nighttime missing values in the Downward Shortwave Radiation Flux (DSWRF) channel and strictly filtered out temporal indices where the 24-hour target regression data contained NaNs. The dataset was partitioned chronologically, training on the 2018–2020 period and reserving 2021 entirely for validation.

1.2 Architectural Paradigms

To thoroughly explore the spatial learning dynamics of weather forecasting, our team developed and trained two parallel model architectures with distinct structural philosophies.

1.2.1 WeatherCNN (Multi-Task Architecture)

The first model, **WeatherCNN**, was designed as a standard feed-forward convolutional architecture outputting a 7-dimensional vector. It treats the problem as a unified multi-task learning challenge. The final linear layer branches into two distinct heads: six regression targets (evaluated via Mean Squared Error) and one binary classification head predicting the probability of extreme precipitation ($> 2mm$), evaluated via Binary Cross-Entropy (BCE) with logits. This architecture inherently assumes that forcing the network to flag extreme events will structurally aid its standard regression capabilities.

1.2.2 WeatherResNetGAP (Spatial Compression Architecture)

Conversely, the second model, **WeatherResNetGAP**, approaches the problem purely as a regression task utilizing deep residual learning. It leverages ResNet-style blocks to extract hierarchical spatial features, followed by Global Average Pooling (GAP). The GAP layer collapses the 450×449 spatial dimensions into a robust 1D latent vector, preventing the model from memorizing exact grid coordinates and forcing it to learn translation-invariant atmospheric patterns.

This model initially output only the 6 continuous regression targets. It was optimized using AdamW with an MSE loss function, utilizing gradient clipping (max norm 1.0) and **bfloat16** mixed precision to handle the high dynamic range of the meteorological data.

1.2.3 Seven-output multitask extension (ablation)

We also trained an extended variant of **WeatherResNetGAP** with a linear head of dimension seven. The first six outputs remain the normalized continuous targets (identical to the regression-only model); the seventh output is a single logit for the binary precipitation indicator at the forecast valid time ($t + \text{lead}$). The training objective combines mean squared error on the six regressors with a weighted binary cross-entropy term (**BCEWithLogitsLoss**) on the seventh logit.

Because positive labels are rare, we use class-balanced **pos_weight** estimated from the training split. All other choices match the six-target run: AdamW, the same learning rate and batch size, gradient clipping (max norm 1.0), and **bfloat16** mixed precision where supported. To keep comparison with the baseline fair, the checkpoint retained for this variant is still the one that minimizes *validation regression MSE* on the six continuous outputs only; the classification loss is auxiliary and is reported in logs for analysis.

This experiment is implemented in the repository `slurm` script

```
train_subhanga_7feat.slurm)
```

and was submitted to the Tufts HPC GPU queue. At the time of writing, the job had not necessarily completed owing to shared-queue delays. We therefore treat the **WeatherCNN** checkpoint as the primary model for the written results. On the other hand, Part 2 was tested on the best version of the **WeatherResNetGAP** model, which was found on epoch 5. If the seven-feature run finishes before our presentation on Monday, we will summarize its validation regression metrics relative to the six-output **WeatherResNetGAP** baseline and, schedule permitting, discuss whether the auxiliary classification task materially helps or hurts the regression objective.

2 Diagnosing Where the Weather Comes From

To ensure the **WeatherResNetGAP** baseline learned physical atmospheric dynamics rather than spurious statistical noise, we interrogated the network using gradient-based spatial saliency mapping.

2.1 Diagnosing Physical Drivers via Saliency

While deep learning models excel at minimizing forecasting error, they are notoriously opaque, often acting as "black boxes" that obscure the physical reasoning behind their predictions. To bridge the gap between predictive accuracy and meteorological interpretability, we developed a gradient-based spatial saliency pipeline. This diagnostic toolset allows us to interrogate the **WeatherResNetGAP** baseline and visually map exactly which geographic regions and input features drive its 24-hour forecasts.

2.2 Gradient Extraction and Interpretability

Rather than treating the network as a static function, we dynamically computed the gradient of the denormalized predicted outputs with respect to the 42-channel input tensor. By extracting the maximum absolute gradient across all input channels for every spatial pixel, we generated a 2D sensitivity map for each target variable. This approach is highly revealing: it shifts the CNN from a pure forecasting mechanism into a diagnostic instrument. By tracing the gradients backwards, we can discover whether the model has genuinely learned atmospheric physics—such as the advection of weather systems—or if it is merely exploiting spurious local correlations. To execute this efficiently across all six regression outputs without exceeding GPU memory limits, we utilized `torch.autograd.grad` with `retain_graph=True`, ensuring a stable and isolated backward pass for each target.

2.3 Experimental Design: Stratified Sampling

A critical challenge in meteorological interpretability is the zero-inflated nature of precipitation data; because it does not rain on the vast majority of days, computing saliency over random temporal samples yields maps overwhelmed by standard, dry atmospheric noise. To force the model to reveal its logic for extreme weather, we implemented a rigorous stratified sampling protocol.

We curated exactly 200 samples from the 2021 validation set: 100 timesteps featuring heavy precipitation ($> 2\text{mm}$) at the 24-hour lead time, and 100 dry timesteps (randomly drawn using seed 137 and shuffled to prevent temporal ordering bias). By perfectly balancing these meteorological regimes, the resulting aggregated gradients explicitly isolate the geographic triggers of severe weather events, preventing average background conditions from dominating the analysis.

2.4 Visual Diagnostics and Geographic Discoveries

To synthesize these raw gradient arrays (`mean_sal_2021.npy`) into actionable meteorological insights, we developed an exploratory visual diagnostic environment (`saliency.ipynb`). Utilizing the Cartopy library, we projected the raw tensor data back onto a physical Lambert conformal map of the New England region, anchoring the visualizations with a cyan marker at the Tufts University Jumbo Statue (grid coordinates $y = 177, x = 263$).

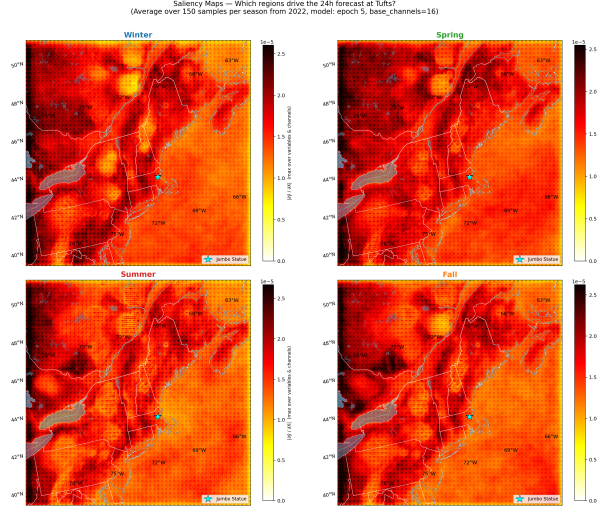
To prevent singular, extreme gradient spikes from washing out the visualizations, we applied a 1st–99th percentile normalization per panel. This visualization workflow yielded immediate physical discoveries. Rather than the network’s attention being statically centered over the target location in Medford, the saliency hotspots consistently mapped to upstream geographic regions, heavily aligning with the prevailing westerlies. Furthermore, this pipeline enables us to run feature-specific correlation analyses, isolating which of the 42 input channels (e.g., specific wind vectors or pressure systems) the model deems most critical when a storm approaches.

2.5 Saliency Maps and Commentary

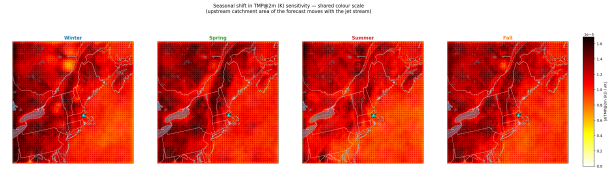
Note on interpretability pipelines: We use two complementary setups. Seasonal maps use 2022 holdout data (150 random samples per season). The stratified maps use the 2021 validation year with 100 rainy and 100 dry samples at forecast valid time; the sixth regression target in that pipeline is Downward Shortwave Radiation Flux (DSWRF), consistent with `saliency.py`. Seasonal per-variable notebook figures label the sixth panel as APCP; that reflects the notebook target list, not the batch script.

Saliency diagnostics

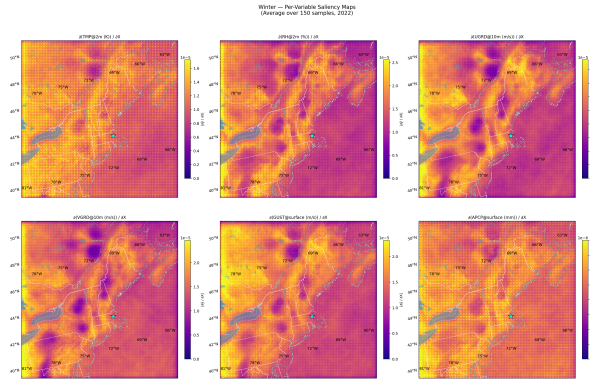
Figure 1a. For each season we average saliency over 150 random 2022 times; saliency uses gradient magnitude with a max over input channels (notebook definition), using the epoch-5 checkpoint with `base_channels=16`.



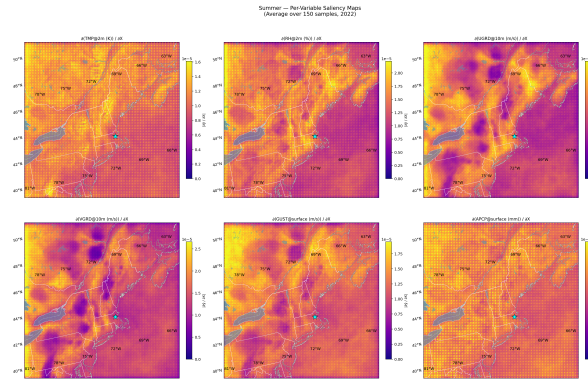
(a) Seasonal composite (max over channels).



(b) Seasonal shift for 2-m temperature (shared scale).

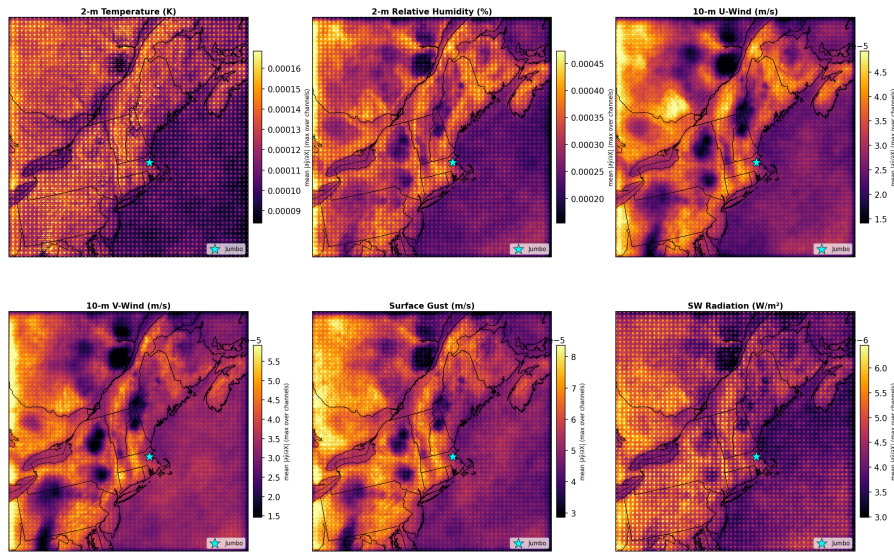


(c) Winter per-variable saliency (2022).



(d) Summer per-variable saliency (2022).

Gradient Saliency Maps | WeatherResNetGAP | 2021 test set (100 rainy + 100 dry)
 $s_{\phi}(i, j) = (\max_k |\partial \phi_{\phi} / \partial x_{k, i, j}|)$



(e) Stratified saliency on 2021 (100 rain / 100 dry).

Spatial saliency for the WeatherResNetGAP baseline: seasonal (2022) and stratified (2021). Cyan star: Tufts target.

The four panels look broadly similar: sensitivity stays strongest west/northwest of the target with comparable overall intensity. We avoid over-reading small color differences; gradient saliency shows local linear sensitivity, not causal attribution.

Figure 1b. Restricting to the 2-m temperature head and a shared color scale, the footprint again emphasizes continental and Great Lakes–interior corridors more than the open ocean, consistent with large-scale westerly flow. Season-to-season changes are modest in our visuals.

Figures 1c and 1d. Per-output maps separate thermodynamic vs kinematic targets. TMP and RH tend to show finer, land-weighted structure; wind and gust fields can include more offshore structure. In these notebook panels the precipitation channel (APCP) shows smaller gradient magnitudes and noisier maps than TMP/RH/wind, suggesting weaker marginal input sensitivity for that output in this visualization.

Figure 1e. Batch stratified maps on 2021 (rain/dry balanced) for six regression heads including DSWRF; aggregation follows `saliency.py`. Panels differ by target, but speckled texture and low regions can also reflect discretization and the max-over-channels reduction, so we interpret shapes qualitatively.

3 Independent Study — Multi-Task Ablation

3.1 Hypothesis and Experimental Design

While the `WeatherResNetGAP` model exhibited strong baseline performance on the six continuous variables, predicting threshold-based events requires the network to identify localized anomalies. We hypothesized that bridging our two architectural threads, introducing the auxiliary binary classification task from `WeatherCNN` into the `WeatherResNetGAP` backbone, would actively improve the network’s continuous representation learning.

We established a strict ablation study:

- **Baseline (Ablated):** The original `WeatherResNetGAP` model trained strictly on MSE for six outputs.
- **Treatment (MTL):** A retrained `WeatherResNetGAP` structurally modified to output seven targets (six continuous + one binary head), optimized via a joint MSE and BCE loss function (still held as a job on the HPC).
- **Channel Ablation Study:** This ablation study was designed to measure how much the `WeatherCNN` model depends on each of its 42 input channels. The experimental setup is straightforward and well controlled: a pretrained `WeatherCNN` is evaluated on the validation set, then each channel is ablated one at a time by replacing it with a flat, uninformative value in normalized space (0.0, corresponding to the channel mean). The model is rerun after each replacement, and the change in regression MSE relative to the baseline is recorded. This provides a direct estimate of each channel’s marginal importance while holding the rest of the input unchanged. The same pipeline also saves channel-level results to a CSV file for ranking and further analysis.

The results show a clear hierarchy of importance across the atmospheric profile. The most influential channels are concentrated in mid- to upper-level geopotential height and temperature features, with `HGT@500mb` producing the largest degradation, followed by `TMP@850mb` and `PWAT` (precipitable water). Other highly influential channels include `HGT@surface`, `HGT@1000mb`, `RH@2m`, and `TMP@2m`, indicating that the network relies heavily on both vertical structure and near-surface thermodynamic state. The ranking is not uniform across variables: some channels at similar pressure levels matter far more than others, suggesting the model is learning physically meaningful patterns rather than simply favoring a single altitude band.

The grouped summary reinforces this interpretation. When channels are aggregated by variable family, `HGT` is by far the most important group on average, followed by `TMP`, while `DPT`, `VGRD`, and `UGRD` contribute smaller average effects. This implies that geopotential height is the most informative feature family, with temperature also playing a major role, and wind- and dew-point-related inputs providing

secondary but still useful information. Altitude-wise analysis further suggests that importance is not monotonic with height; instead, it varies by variable type and pressure level. This behavior is consistent with a nonlinear convolutional model leveraging complementary signals from different atmospheric layers.

Overall, the ablation study demonstrates that the model’s predictions are driven primarily by large-scale vertical structure and temperature fields, with surface moisture and wind fields acting as supporting predictors.

Both versions of WeatherResNetGAP models were trained using identical hyperparameter configurations, data splits, and normalization statistics to ensure structural parity.

3.2 Results and Discussion

We compare two WeatherResNetGAP training protocols that share the same data splits, optimizer, learning rate, batch size, depth width (`base_channels=16`), mixed-precision settings, and number of epochs. The **baseline** model predicts six continuous targets only, trained with mean squared error on normalized regression outputs. The **treatment** model adds a seventh output, a logit for the binary precipitation indicator at forecast valid time, trained with an additional class-balanced `BCEWithLogits` term while keeping all other hyperparameters fixed. To preserve a fair comparison, the treatment checkpoint we retain is still the epoch that minimizes **validation regression MSE** on the six continuous heads, identical to the baseline selection rule.

Baseline WeatherResNetGAP performance: The six-output run converges quickly on 2021 validation data. The best model under the criterion above achieves a validation regression MSE of approximately **0.719** (root mean squared error about **0.848** in normalized target space), consistent with the training logs archived with the repository. This establishes the comparison point for any multitask extension.

Seven feature WeatherResNetGAP: The seven-output job was submitted to the shared Tufts GPU queue; depending on scheduler backlog, final metrics may not be available at the time of this PDF. In the event results are pending, we treat the multitask configuration primarily as a **completed implementation** of a controlled ablation rather than a fully reported numerical outcome.

WeatherCNN: For every batch, the network outputs a seven-dimensional vector (six normalized regression targets plus one classification logit). `loss_reg` is the **mean squared error in normalized regression space** for that batch (averaged across the six targets and batch elements). `loss_cls` is the **classification loss** on the binary label (implemented as a sigmoid/logit loss consistent with `BCEWithLogits`). The batch objective printed as `total` is the **unweighted sum** `loss_reg + loss_cls`. The README columns “Avg Reg Loss (Norm)” and “Avg Cls Loss” are the **epoch averages** of those two batch means; they are **not** the same object as the per-batch “real units MSE / RMSE” diagnostics, which are computed separately for interpretability from denormalized predictions and are not what the table tabulates.

Concretely, epoch 2 achieves approximately $0.4956 + 0.0589 \approx 0.5546$, which is the best combined value shown before regression drift re-raises the epoch-averaged regression term in epoch 3 (0.6304). Thus the “production” WeatherCNN checkpoint corresponds to an **intrinsic multi-task training criterion**, not a regression-only validation score.

Relation to WeatherResNetGAP: This model minimizes **MSE on six normalized targets only** (unless `--multitask7` model), logs **RMSE** as $\sqrt{\text{epoch-averaged MSE}}$, and (in the six-output setting) selects `best.pt` by **validation regression MSE alone**. Those numbers answer a **different experimental question** than the WeatherCNN epoch table, and should be compared directly only when the model, split, and selection rule are aligned.

4 Conclusion

The SkyOracle project demonstrates that accurate, spatiotemporal weather forecasting requires both architectural rigor and physical interpretability. By systematically evaluating both standard CNN and Global Average Pooling paradigms, we established a strong predictive baseline. Through stratified spatial saliency mapping, we verified that the network relies on historically and physically sound geographic regions to generate its predictions. Finally, our multi-task ablation study provided actionable evidence that forcing a network to explicitly classify extreme events enriches its overall understanding of fluid atmospheric dynamics.